

What is Pigeonhole Principle?

- ▶ This principle states that:

“Suppose a flock of pigeons fly into a set of pigeonholes to roost. If there are more pigeons than pigeonholes, then there must be at least 1 pigeonhole that has more than one pigeon in it.”

- ▶ If $k+1$ or more objects are placed into k boxes, then there is at least one box containing two or more of the objects
 - ▶ This is Theorem 1

What is Pigeonhole Principle?



● If $n > m$
pigeons are put
into m
pigeonholes, th
ere's a hole
with more than
one pigeon.

What is Pigeonhole Principle?

- ▶ This principle is believed to have been made by Dirichlet in 1834, under the name of “**Drawer Principle**” or “**Shelf Principle**”.
- ▶ Thus it is also known as **Dirichlet Box (or drawer) principle**.

Pigeonhole principle examples in Counting Problems

- ▶ In a group of 367 people, there must be two people with the same birthday
 - ▶ As there are 366 possible birthdays
- ▶ In a group of 27 English words, at least two words must start with the same letter
 - ▶ As there are only 26 letters

Example 2

- In the morning, John draws socks randomly from his drawer. If there are 12 pairs of socks, each pair a different colour, in the drawer, how many socks does John have to draw at most in order to get a matched pair?

[Sol'n]

There are 12 different colours (pigeonholes) and in order to ensure at least two socks (pigeons) have the same colour,

no. of socks drawn > 12 .

So at most $12+1 = 13$ socks have to be drawn to get a matched pair.

Note

We only know that at most 13 socks are required, but we do not know which colour would the matched pair be.

Generalized pigeonhole principle

- ▶ If N objects are placed into k boxes, then there is at least one box containing $\lceil N/k \rceil$ objects
 - ▶ This is Theorem 2

Generalized pigeonhole principle examples

- ▶ Among 100 people, there are at least $\lceil 100/12 \rceil = 9$ born on the same month
- ▶ How many students in a class must there be to ensure that 6 students get the same grade (one of A, B, C, D, or F)?
 - ▶ The "boxes" are the grades. Thus, $k = 5$
 - ▶ Thus, we set $\lceil N/5 \rceil = 6$
 - ▶ Lowest possible value for N is 26

Example of Generalized Pigeonhole Principle.

- Show that if 9 colours are used to paint 100 houses, then at least 12 houses will be of the same colour.

[Sol'n]

We have 9 colours (pigeonholes) and 100 houses (pigeons). By the generalized pigeonhole principle, there is at least one colour that will be used to paint at least $\left\lceil \frac{100}{9} \right\rceil = 12$ houses.

So there is at least 12 houses painted with the same colour.

Implementation of Pigeonhole Principles

- Pigeonhole principle is applied in the following problems that fall under following areas:
- Problems on functions
- Problems on relations
- Problems on divisibility
- Problems on geometry
- Problems on probability theory